

PFIZER
CIBINQO™*

1. NAME OF THE MEDICINAL PRODUCT

CIBINQO 50 mg film-coated tablets
CIBINQO 100 mg film-coated tablets
CIBINQO 200 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

CIBINQO 50 mg film-coated tablets

Each film-coated tablet contains 50 mg abrocitinib

CIBINQO 100 mg film-coated tablets

Each film-coated tablet contains 100 mg abrocitinib

CIBINQO 200 mg film-coated tablets

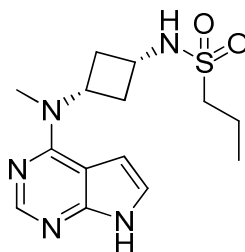
Each film-coated tablet contains 200 mg abrocitinib

Excipients with known effects

Each CIBINQO 50 mg film-coated tablet contains 1.365 mg of lactose monohydrate.
Each CIBINQO 100 mg film-coated tablet contains 2.73 mg of lactose monohydrate.
Each CIBINQO 200 mg film-coated tablet contains 5.46 mg of lactose monohydrate.

For a full list of excipients, see Section 6.1

Structure of abrocitinib:



3. PHARMACEUTICAL FORM

Film-coated tablet.

CIBINQO 50 mg film-coated tablets

Pink, oval tablet 11 mm long and 5 mm wide, debossed with “PFE” on one side and “ABR 50” on the other.

CIBINQO 100 mg film-coated tablets

Pink, round tablet 9 mm in diameter, debossed with “PFE” on one side and “ABR 100” on the other.

CIBINQO 200 mg film-coated tablets

Pink, oval tablet 18 mm long and 8 mm wide debossed with “PFE” on one side and “ABR 200” on the other.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

CIBINQO is indicated for the treatment of patients 12 years of age and older with moderate to severe atopic dermatitis, including the relief of pruritus, who have had an inadequate response to prescribed topical therapy or for whom these treatments are not advisable. CIBINQO can be used with or without medicated topical therapies for atopic dermatitis.

4.2 Posology and Method of Administration

Posology

- The recommended starting dose of CIBINQO is 100 mg or 200 mg once daily based on individual patient characteristics: A starting dose of 100 mg once daily is recommended for patients at higher risk of venous thromboembolism (VTE), major adverse cardiovascular event (MACE) and malignancy (see Section 4.4). If the patient does not respond adequately to 100 mg once daily, the dose can be increased to 200 mg once daily (see below).
- A dose of 200 mg once daily may be appropriate for patients who are not at higher risk of VTE, MACE and malignancy or for patients with an inadequate response to 100 mg once daily. Upon disease control, dose should be decreased to 100 mg once daily. If disease control is not maintained after dose reduction, re-treatment with 200 mg once daily can be considered.

The lowest effective dose for maintenance should be considered. Discontinuation of treatment should be considered in patients who show no evidence of therapeutic benefit after 24 weeks.

CIBINQO can be used with or without medicated topical therapies for atopic dermatitis.

Treatment initiation

Treatment with CIBINQO should not be initiated in patients with a platelet count $<150 \times 10^3/\text{mm}^3$, an absolute lymphocyte count (ALC) $<0.5 \times 10^3/\text{mm}^3$, an absolute neutrophil count (ANC) $<1 \times 10^3/\text{mm}^3$ or who have a hemoglobin value $<8 \text{ g/dL}$ (see Section 4.4).

Dose interruption

If a patient develops a serious infection, sepsis or opportunistic infection, consider interruption of CIBINQO until the infection is controlled should be considered (see Section 4.4).

Interruption of dosing may be needed for management of laboratory abnormalities as described in Table 1 (see Section 4.4).

Missed doses

If a dose is missed, patients should be advised to take the dose as soon as possible unless it is less than 12 hours before the next dose, in which case the patient should not take the missed dose. Thereafter, resume dosing at the regular scheduled time.

Drug-drug interactions

When recommended dose is 100 mg or 200 mg CIBINQO once daily, dose should be reduced by 50% to 50 mg or 100 mg once daily, respectively, in patients taking strong inhibitors of cytochrome P450 (CYP) 2C19 (e.g., fluvoxamine, fluconazole). The use of CIBINQO is not recommended concomitantly with moderate or strong inducers of CYP2C19/CYP2C9 enzymes (e.g., rifampin, apalutamide, efavirenz, enzalutamide, phenytoin) (see Section 4.5).

Special populations

Renal impairment

No dose adjustment is required in patients with mild renal impairment, i.e., estimated glomerular filtration rate (eGFR) of 60 to <90 mL/min.

In patients with moderate (eGFR 30 to <60 mL/min) renal impairment, the recommended dose of CIBINQO should be reduced by half to 100 mg or 50 mg once daily (see Section 5.2).

In patients with severe (eGFR <30 mL/min) renal impairment, 50 mg once daily is the recommended starting dose. The maximum daily dose is 100 mg (see Section 5.2).

Hepatic impairment

No dose adjustment is required in patients with mild (Child Pugh A) or moderate (Child Pugh B) hepatic impairment. CIBINQO has not been studied in patients with severe (Child Pugh C) hepatic impairment (see Section 5.2).

Elderly population

The recommended starting dose for patients ≥ 65 years of age is 100 mg once daily (see Section 4.4).

Pediatric population

The safety and efficacy of CIBINQO in pediatric patients under 12 years of age have not yet been established. No data are available.

Method of administration

CIBINQO is to be taken orally once daily with or without food at approximately the same time each day.

In patients who experience nausea, taking CIBINQO with food may improve nausea.

Swallow CIBINQO tablets whole and intact with water. Do not crush, split, or chew CIBINQO tablets.

4.3 Contraindications

None

4.4 Special Warnings and Precautions for Use

Infections/serious infections

Serious infections have been reported in patients receiving CIBINQO. The most frequent serious infections in clinical studies were herpes simplex, herpes zoster, and pneumonia (see Section 4.8).

The risks and benefits of treatment with CIBINQO should be carefully considered for patients.

- with chronic or recurrent infection
- who have been exposed to TB
- with a history of a serious or an opportunistic infection
- who have resided or travelled in areas of endemic TB or endemic mycoses; or

- with underlying conditions that may predispose them to infection.

Patients should be closely monitored for the development of signs and symptoms of infection during and after treatment with CIBINQO. A patient who develops a new infection during treatment with CIBINQO should undergo prompt and complete diagnostic testing and appropriate antimicrobial therapy should be initiated. The patient should be closely monitored and CIBINQO therapy should be interrupted if the patient is not responding to standard therapy.

Tuberculosis

Tuberculosis was observed in clinical studies with abrocitinib. Patients should be screened for tuberculosis (TB) before starting CIBINQO therapy and consider yearly screening for patients in highly endemic areas for TB. CIBINQO should not be given to patients with active TB. For patients with a new diagnosis of latent TB or prior untreated latent TB, preventive therapy for latent TB should be started prior to initiation of CIBINQO.

Viral reactivation

Viral reactivation, including herpes virus reactivation (e.g., herpes zoster, herpes simplex), was reported in clinical studies (see Section 4.8). The rate of herpes zoster infections was higher in patients who were treated with 200 mg, 65 years of age and older, with a medical history of herpes zoster, with a confirmed ALC $<1 \times 10^3/\text{mm}^3$ prior to the event and patients with severe atopic dermatitis at baseline (see Section 4.8). If a patient develops herpes zoster, temporary interruption of treatment may be considered until the episode resolves.

Screening for viral hepatitis should be performed in accordance with clinical guidelines before starting therapy and during therapy with CIBINQO. Patients with evidence of active hepatitis B or hepatitis C (positive hepatitis C PCR) infection were excluded from clinical studies (see Section 5.2). Patients who were hepatitis B surface antigen negative, hepatitis B core antibody positive, and hepatitis B surface antibody positive had testing for hepatitis B virus (HBV) DNA. Patients who had HBV DNA above the lower limit of quantification (LLQ) were excluded. Patients who had HBV DNA negative or below LLQ could initiate treatment with CIBINQO; such patients had HBV DNA monitored. If HBV DNA is detected, a liver specialist should be consulted.

Vaccination

No data are available on the response to vaccination in patients receiving CIBINQO. Use of live, attenuated vaccines should be avoided during or immediately prior to treatment. Prior to initiating CIBINQO, it is recommended that patients be brought up to date with all immunizations, including prophylactic herpes zoster vaccinations, in agreement with current immunization guidelines.

Venous thromboembolism (VTE)

Events of deep venous thrombosis (DVT) and pulmonary embolism (PE) have been reported in patients receiving Janus kinase (JAK) inhibitors including abrocitinib (see Section 4.8).

In a large randomized active-controlled study of tofacitinib (another JAK inhibitor) in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, a dose-dependent higher rate of VTE including deep venous thrombosis (DVT) and pulmonary embolism (PE) was observed with tofacitinib compared to TNF inhibitors.

CIBINQO should be used with caution in patients at risk for DVT/PE. Risk factors that should be considered in determining the patient's risk for DVT/PE include age ≥ 65 years, long-term smoking, current malignancy (excluding non-melanoma skin cancer [NMSC]), a

medical history of DVT/PE, prothrombotic disorder, use of combined hormonal contraceptives or hormone replacement therapy, patients undergoing major surgery, or prolonged immobilization. Patients should be re-evaluated periodically during abrocitinib treatment to assess for changes in VTE risk.

Promptly evaluate patients with signs and symptoms of VTE and discontinue abrocitinib in patients with suspected VTE, regardless of dose.

Major adverse cardiovascular events (MACE)

Events of MACE have been observed in patients taking abrocitinib.

In a large randomized active-controlled study of tofacitinib (another JAK inhibitor) in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, a higher rate of major adverse cardiovascular events (MACE), defined as cardiovascular death, non-fatal myocardial infarction (MI) and non-fatal stroke, was observed with tofacitinib compared to TNF inhibitors.

Therefore, in patients 65 years of age and older, patients who are long-term current or long-term past smokers, and patients with a history of atherosclerotic cardiovascular disease or other cardiovascular risk factors, abrocitinib should be used with caution.

Malignancy (excluding non-melanoma skin cancer [NMSC])

Lymphoma and other malignancies have been reported in patients receiving JAK inhibitors, including abrocitinib.

In a large randomized active-controlled study of tofacitinib (another JAK inhibitor) in rheumatoid arthritis patients 50 years and older with at least one additional cardiovascular risk factor, a higher rate of malignancies, particularly lung cancer, lymphoma and non-melanoma skin cancer (NMSC) was observed with tofacitinib compared to TNF inhibitors.

In patients 65 years of age and older, patients who are long-term current or long-term past smokers, or with other malignancy risk factors (e.g., current malignancy or history of malignancy), abrocitinib should be used with caution.

Non-melanoma skin cancer

NMSCs have been reported in patients receiving abrocitinib. Periodic skin examination is recommended for all patients, particularly those who are at increased risk for skin cancer.

Hematologic abnormalities

Confirmed ALC $<0.5 \times 10^3/\text{mm}^3$ and platelet count $<50 \times 10^3/\text{mm}^3$ were observed in less than 0.5% of patients in clinical studies (see Section 4.8). Treatment with CIBINQO should not be initiated in patients with a platelet count $<150 \times 10^3/\text{mm}^3$, an ALC $<0.5 \times 10^3/\text{mm}^3$, an ANC $<1 \times 10^3/\text{mm}^3$ or who have a hemoglobin value <8 g/dL (see Section 4.2). Complete blood count should be monitored 4 weeks after initiation of therapy with CIBINQO and thereafter according to routine patient management (see Table 1).

Lipids

Dose-dependent increase in blood lipid parameters were reported in patients treated with CIBINQO (see Section 4.8). Lipid parameters should be assessed approximately 4 weeks following initiation of CIBINQO therapy and thereafter according to their risk for cardiovascular disease (see Table 1). The effect of these lipid parameter elevations on cardiovascular morbidity and mortality has not been determined. Patients with abnormal lipid

parameters should be further monitored and managed according to clinical guidelines, due to the known cardiovascular risks associated with hyperlipidemia. If abrocitinib is chosen interventions to manage lipid concentrations should be implemented according to clinical guidelines.

Laboratory monitoring

Table 1. Laboratory monitoring guidance

Laboratory measures	Monitoring guidance	Action
Complete blood count including Platelet Count, Absolute Lymphocyte Count (ALC), Absolute Neutrophil Count (ANC) and Haemoglobin (Hb)	Before treatment initiation, 4 weeks after initiation and thereafter according to routine patient management.	Platelets: Treatment should be discontinued if platelet counts are $<50 \times 10^3/\text{mm}^3$
		ALC: Treatment should be interrupted if ALC is $<0.5 \times 10^3/\text{mm}^3$ and may be restarted once ALC returns above this value. Treatment should be discontinued if confirmed.
		ANC: Treatment should be interrupted if ANC is $<1 \times 10^3/\text{mm}^3$ and may be restarted once ANC returns above this value.
		Hb: Treatment should be interrupted if Hb is $<8 \text{ g/dL}$ and may be restarted once Hb returns above this value.
Lipid parameters	Before treatment initiation, 4 weeks after initiation and thereafter according to the patient's risk for cardiovascular disease and clinical guidelines for hyperlipidaemia.	Patients should be monitored according to clinical guidelines for hyperlipidaemia.

Elderly

A total of 176 patients 65 years of age and older were enrolled in CIBINQO studies. The safety profile observed in elderly patients was similar to that of the adult population with the following exceptions: a higher proportion of patients 65 years of age and older discontinued from clinical studies and were more likely to have serious adverse events compared to younger patients; patients 65 years and older were more likely to develop low platelet and ALC values; the incidence rate of herpes zoster in patients 65 years of age and older was higher than that of younger patients (see Section 4.8). There are limited data in patients above 75 years of age.

Use in patients 65 years of age and older

Considering the increased risk of MACE, malignancies, serious infections, and all-cause mortality in patients 65 years of age and older, as observed in a large randomized study of tofacitinib (another JAK inhibitor), abrocitinib should be used with caution in these patients.

Immunosuppressive conditions or medicinal products

Patients with immunodeficiency disorders or a first-degree relative with a hereditary immunodeficiency were excluded from clinical studies and no information on these patients is available.

Combination with biologic immunomodulators, potent immunosuppressants such as

ciclosporin or other Janus kinase (JAK) inhibitors has not been studied. Their concomitant use with abrocitinib is not recommended as a risk of additive immunosuppression cannot be excluded.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Potential for other medicines to affect pharmacokinetics of CIBINQO

Abrocitinib is metabolized predominantly by CYP2C19 and CYP2C9 enzymes and to a lesser extent by CYP3A4 and CYP2B6 enzymes, and its active metabolites are renally excreted and are substrates of the organic anion transporter 3 (OAT3). Therefore, exposures of abrocitinib and/or its active metabolites may be affected by medicinal products that strongly inhibit or induce CYP2C19 or CYP2C9 or inhibit the OAT3 transporter. Dose adjustments, as appropriate, based on these results are outlined in Section 4.2.

Coadministration with CYP2C19/CYP2C9 inhibitors

When CIBINQO 100 mg was administered concomitantly with fluvoxamine (a strong CYP2C19 and moderate CYP3A inhibitor) or fluconazole (a strong CYP2C19, moderate CYP2C9 and CYP3A inhibitor), the extent of exposure of abrocitinib active moiety increased by 91% and 155%, respectively, compared with administration alone.

Coadministration with CYP2C19/CYP2C9 inducers

Administration of CIBINQO 200 mg after multiple dosing with rifampin, a strong inducer of CYP enzymes, resulted in reduction of abrocitinib active moiety exposures by approximately 56%. Based on the results of PBPK analysis, moderate induction of CYP enzymes reduces the exposure of abrocitinib active moiety by 44%.

Coadministration with OAT3 inhibitors

When CIBINQO 200 mg was administered concomitantly with probenecid, an OAT3 inhibitor, abrocitinib active moiety exposures increased by approximately 66%. This is not clinically significant, and a dose adjustment is not needed.

Coadministration with products which increase gastric pH

The effect of elevating gastric pH on abrocitinib active moiety exposures is not clinically significant and dose adjustment is not needed.

When abrocitinib 200 mg was administered concomitantly with famotidine 40 mg, an H₂-receptor antagonist, the peak and extent of abrocitinib active moiety exposures decreased by approximately 82% and 20% respectively. The effect of elevating gastric pH with antacids, or proton pump inhibitors (omeprazole) on the pharmacokinetics of abrocitinib has not been studied and may reduce the absorption of abrocitinib in a manner similar to that seen with famotidine.

Potential for CIBINQO to affect pharmacokinetics of other medicines

No clinically significant effects of CIBINQO were observed in drug interaction studies with oral contraceptives (e.g., ethinyl estradiol/levonorgestrel), or with substrates of BCRP and OAT3 (e.g., rosuvastatin), MATE1/2K (e.g., metformin), CYP3A4 (e.g., midazolam), CYP1A2 (e.g., caffeine) and CYP2B6 (e.g., efavirenz). *In vitro* abrocitinib is an inhibitor of P-glycoprotein (P-gp). Coadministration of dabigatran etexilate (a P-gp substrate), with a single dose of CIBINQO 200 mg increased dabigatran AUC_{inf} and C_{max} by approximately 53% and 40%, respectively, compared with administration alone. Caution should be exercised for concomitant use of abrocitinib with dabigatran. The effect of abrocitinib on the pharmacokinetics of other P-gp substrates has not been evaluated. Caution should be exercised as the levels of P-gp substrates with a narrow therapeutic index, such as digoxin, may increase.

In vitro, abrocitinib is an inhibitor of CYP2C19 enzyme. Coadministration of abrocitinib 200 mg once daily with omeprazole 10 mg single dose increased the AUC_{inf} and C_{max} of omeprazole by approximately 189% and 134%, respectively, indicating that abrocitinib is a moderate inhibitor of CYP2C19 enzyme. Caution should be exercised when using abrocitinib concomitantly with narrow therapeutic index medicines that are primarily metabolized by CYP2C19 enzyme (e.g., S-mephenytoin, clopidogrel).

4.6 Fertility, Pregnancy and Lactation

Women of childbearing potential

Women of reproductive potential should be advised to use effective contraception during treatment and for 1 month following the final dose of CIBINQO. Consider pregnancy planning and prevention for females of reproductive potential.

Pregnancy

The limited human data on use of abrocitinib in pregnant women are not sufficient to evaluate a drug-associated risk for major birth defects or miscarriage. In embryo-fetal development studies, oral administration of abrocitinib to pregnant rats during organogenesis resulted in fetotoxicity at exposures equal to 17 times the unbound AUC at the maximum recommended human dose (MRHD) of 200 mg once daily. In embryo-fetal development studies, oral administration of abrocitinib to pregnant rabbits did not result in fetotoxicity at exposures equal to 4 times the unbound AUC at the MRHD of 200 mg. In a pre- and postnatal development study in pregnant rats, abrocitinib oral administration during gestation and through lactation resulted in lower postnatal survival and lower offspring body weights at exposures greater than or equal to 11 times the unbound AUC at the MRHD of 200 mg once daily (see Section 5.3). CIBINQO should not be used during pregnancy unless clearly necessary.

Breastfeeding

There are no data on the presence of abrocitinib in human milk, the effects on the breastfed infant, or the effects on milk production. Abrocitinib was secreted in milk of lactating rats. A risk to newborns/infants cannot be excluded and CIBINQO should not be used during breastfeeding.

Fertility

Based on the findings in rats, oral administration of CIBINQO may result in temporary reduced fertility in females of reproductive potential. These effects on female rat fertility were reversible 1 month after cessation of CIBINQO oral administration (see Section 5.3).

4.7 Effects on Ability to Drive and Use Machines

CIBINQO has no or negligible sedating effect. However, patients who experience dizziness after the intake of abrocitinib should refrain from driving or using machines until the dizziness resolves.

4.8 Undesirable Effects

Summary of safety profile

The most commonly reported adverse reactions (ARs) occurring in $\geq 2\%$ of patients treated with CIBINQO in placebo-controlled studies were nausea (15.1%), headache (7.9%), acne (4.8%), herpes simplex (4.2%), vomiting (3.5%), dizziness (3.4%), blood creatine phosphokinase increased (3.1%), and abdominal pain upper (2.2%). The most frequent

serious adverse reactions in atopic dermatitis patients were infections (see Section 4.4).

Tabulated list of adverse reactions

A total of 3848 patients were treated with CIBINQO in clinical studies in atopic dermatitis; among them 3050 patients (representing 5166 patient-years of exposure) were integrated for safety analysis, 2013 patients with at least 48 weeks of exposure. The integrated safety analysis included 1997 patients receiving a constant dose of abrocitinib 200 mg and 1053 patients receiving a constant dose of 100 mg. Five placebo-controlled studies were integrated (703 patients on 100 mg once daily, 684 patients on 200 mg once daily and 438 patients on placebo) to evaluate the safety of CIBINQO in comparison to placebo for up to 16 weeks.

Table 2. Adverse Drug Reactions (ADRs) by System Organ Class and Council for International Organizations of Medical Science (CIOMS) Frequency Category Listed in Order of Decreasing Medical Seriousness or Clinical Importance Within Each Frequency Category and SOC

System Organ Class	Very Common ≥1/10	Common ≥1/100 to <1/10	Uncommon ≥1/1,000 to <1/100
Infections and infestations		Herpes simplex ^a Herpes zoster ^b	Pneumonia
Blood and lymphatic system disorders		Thrombocytopenia	Lymphopenia
Metabolism and nutrition disorders			Hyperlipidaemia ^c
Nervous system disorders		Headache Dizziness	
Vascular disorders			Venous thromboembolism ^d .
Gastrointestinal disorders	Nausea	Vomiting Abdominal pain upper	
Skin and subcutaneous tissue disorders		Acne	
Investigations		Blood creatine phosphokinase increased	

a. Herpes simplex includes oral herpes, ophthalmic herpes simplex, genital herpes, and herpes dermatitis.

b. Herpes zoster includes ophthalmic herpes zoster.

c. Hyperlipidaemia included dyslipidaemia and hypercholesterolaemia.

d. Venous thromboembolism includes pulmonary embolism and deep vein thrombosis.

Description of selected adverse reactions

Infections

In placebo-controlled studies, for up to 16 weeks, overall infections have been reported in 27.4% of patients treated with placebo and in 34.9% and 34.8% of patients treated with CIBINQO 100 mg and 200 mg, respectively. Most infections were mild or moderate.

The percentage of patients reporting infection-related adverse reactions in the 200 mg and 100 mg groups compared to placebo were: herpes simplex (4.2% and 2.8% versus 1.4%), herpes zoster (1.2% and 0.6% versus 0%), pneumonia (0.1% and 0.1% versus 0%). Herpes simplex was more frequent in patients with a history of herpes simplex or eczema herpeticum. Most of the herpes zoster events involved a single dermatome and were non-serious.

Among all patients treated in clinical studies with consistent dosing regimens of either CIBINQO 100 mg or 200 mg, including the long-term extension study, the incidence rate of herpes zoster in patients treated with abrocitinib 200 mg (4.36 per 100 patient-years) was

higher than that of patients treated with 100 mg (2.61 per 100 patient-years). Incidence rates for herpes zoster were also higher for patients 65 years of age and older (HR 1.76), patients with a medical history of herpes zoster (HR 3.41), patients with severe atopic dermatitis at baseline (HR 1.17), and a confirmed ALC $<1.0 \times 10^3/\text{mm}^3$ prior to the event of herpes zoster (HR 2.18) (see Section 4.4).

In placebo-controlled studies, for up to 16 weeks, the rate of serious infections was 1.81 per 100 patient-years in patients treated with placebo, 3.32 per 100 patient-years in patients treated with 100 mg, and 1.12 per 100 patient-years in patients treated with 200 mg. Among all patients treated in clinical studies with consistent dosing regimens of either CIBINQO 100 mg or 200 mg, including the long-term extension study, the rate of serious infections was 2.20 per 100 patient-years in the CIBINQO 100 mg group and 2.48 per 100 patient-years in the CIBINQO 200 mg group. The most commonly reported serious infections were herpes simplex, herpes zoster, and pneumonia (see Section 4.4).

Opportunistic infections

Most opportunistic infections were non-serious cases of multidermatomal cutaneous herpes zoster. Among all patients treated in clinical studies with consistent dosing regimens of either CIBINQO 100 mg or 200 mg, including the long-term extension study, the rate of opportunistic infections was 0.70 per 100 patient-years in the CIBINQO 100 mg group and 0.96 per 100 patient-years in the CIBINQO 200 mg group. Most cases of opportunistic herpes zoster were mild or moderate.

Venous thromboembolism

Among all patients treated in clinical studies with consistent dosing regimen of either CIBINQO 100 mg or 200 mg, including the long-term extension study, the rate of PE was 0.05 per 100 patient-years in the CIBINQO 100 mg group and 0.21 per 100 patient-years in the CIBINQO 200 mg group. The rate of DVT was 0.05 per 100 patient-years in the CIBINQO 100 mg group and 0.06 per 100 patient years in the CIBINQO 200 mg group (see Section 4.4).

Thrombocytopenia

In placebo-controlled studies, for up to 16 weeks, treatment with CIBINQO was associated with a dose-related decrease in platelet count. Maximum effects on platelets were observed within 4 weeks, after which the platelet count returned towards baseline despite continued therapy. Confirmed platelet counts of $<50 \times 10^3/\text{mm}^3$ were reported in 0.1% of patients exposed to CIBINQO 200 mg, 0 patients treated with CIBINQO 100 mg or placebo. Among all patients treated in clinical studies with consistent dosing regimens of either CIBINQO 100 mg or 200 mg, including the long-term extension study, the rate of confirmed platelet counts of $<50 \times 10^3/\text{mm}^3$ was 0.15 per 100 patient-years for 200 mg and 0 per 100 patient-years for 100 mg, most occurring at Week 4. Patients 65 years of age and older had a higher rate of platelet counts $<75 \times 10^3/\text{mm}^3$ (see Section 4.4). There were no adolescent patients who developed platelet counts $<75 \times 10^3/\text{mm}^3$.

Lymphopenia

In placebo-controlled studies, for up to 16 weeks, confirmed ALC $<0.5 \times 10^3/\text{mm}^3$ occurred in 2 patients (0.3%) treated with CIBINQO 200 mg and 0 patients treated with CIBINQO 100 mg or placebo. Both cases occurred in the first 4 weeks of exposure. Among all patients treated in clinical studies with consistent dosing regimens of either CIBINQO 100 mg or 200 mg, including the long-term extension, the rate of confirmed ALC $<0.5 \times 10^3/\text{mm}^3$ was 0.34 per 100 patient-years for 200 mg and 0.05 per 100 patient-years for 100 mg, the highest rate was observed in patients 65 years of age and older (see Section 4.4). There were no adolescent patients who developed an ALC $<0.5 \times 10^3/\text{mm}^3$.

Lipid elevations

In placebo-controlled studies, for up to 16 weeks, there was a dose-related percent increase in low-density lipoprotein cholesterol (LDL-c), total cholesterol, and high-density lipoprotein

cholesterol (HDL-c) relative to placebo at Week 4 which remained elevated through the final visit in the treatment period. The median % change in LDL-c at Week 4 was 9.1%, 4.9% and -2.8% in patients exposed to 200 mg, 100 mg and placebo, respectively. The median % change in HDL-c at Week 4 was 20.0%, 12.1%, and 0% in patients exposed to 200 mg, 100 mg and placebo, respectively. Events related to hyperlipidemia occurred in 0.4% of patients exposed to CIBINQO 100 mg, 0.6% of patients exposed to 200 mg, and 0% of patients exposed to placebo (see Section 4.4).

Creatine phosphokinase elevations (CPK)

In placebo-controlled studies, for up to 16 weeks, events of blood CPK increased ($>5 \times \text{ULN}$) were reported in 3.8% of patients treated with 200 mg, 1.8% of patients treated with 100 mg of CIBINQO and 1.8% of patients treated with placebo. Most elevations were transient, and none led to discontinuation.

Nausea

Nausea was most frequent in the first week of CIBINQO therapy and generally resolved with continued therapy. The median duration of nausea was 15 days. Most of the cases were mild to moderate in severity.

Pediatric population

A total of 635 adolescent patients (12 to less than 18 years of age) were treated with abrocitinib in clinical studies in atopic dermatitis representing 1326.1 patient-years of exposure. The safety profile observed in adolescents in atopic dermatitis clinical studies was similar to that of the adult population.

4.9 Overdose

CIBINQO was administered in clinical studies up to a single oral dose of 800 mg. There is no experience with overdose of CIBINQO. There is no specific antidote for overdose with CIBINQO. In case of an overdose, it is recommended that the patient be monitored for signs and symptoms of adverse reactions. Treatment should be symptomatic and supportive.

Pharmacokinetic data up to and including a single oral dose of 800 mg in healthy adult volunteers indicate that more than 90% of the administered dose is expected to be eliminated within 48 hours.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Mechanism of action

CIBINQO is a Janus kinase (JAK) 1 inhibitor. JAKs are intracellular enzymes which transmit signals arising from cytokine or growth factor-receptor interactions on the cellular membrane to influence cellular processes of hematopoiesis and immune cell function. JAKs phosphorylate and activate Signal Transducers and Activators of Transcription (STATs) which modulate intracellular activity including gene expression. Inhibition of JAK1 modulates the signaling pathway by preventing the phosphorylation and activation of STATs.

In biochemical assay, abrocitinib has selectivity for JAK1 over the other 3 JAK isoforms JAK2 (28-fold), JAK3 (>340 -fold) and tyrosine kinase 2 (TYK2, 43-fold). In cellular settings, it preferentially inhibits cytokine induced STAT phosphorylation by signaling pairs involving JAK1 and spares signaling by JAK2/JAK2 or JAK2/TYK2 pairs. The relevance of selective enzymatic inhibition of specific JAK enzymes to clinical effect is not currently known.

Pharmacodynamic effects

Treatment with CIBINQO was associated with dose-dependent reduction in serum markers of inflammation, including high sensitivity C-reactive protein (hsCRP), interleukin-31 (IL-31) and thymus and activation-regulated chemokine (TARC). These changes returned to near baseline within 4 weeks of drug discontinuation.

Mean absolute lymphocyte count increased by 2 weeks after starting treatment with abrocitinib and returned to baseline by Month 9 of treatment. Most patients maintained an ALC within the reference range. Treatment with abrocitinib was associated with a dose-related increase in B cell counts and a dose-related decrease in NK cell counts. The clinical significance of these changes in B cell and NK cell counts is unknown.

Clinical efficacy and safety

The efficacy and safety of CIBINQO as monotherapy and in combination with background medicated topical therapies over 12-16 weeks were evaluated in 1616 patients in 3 pivotal randomized, double-blind, placebo-controlled studies (MONO-1, MONO-2, and COMPARE). In addition, the efficacy and safety of CIBINQO in monotherapy over 52 weeks (with the option of rescue treatment in flaring patients) was evaluated in 1233 patients in a Phase 3 induction, randomized withdrawal, double-blind, placebo-controlled study (REGIMEN). The patients in these 4 studies were 12 years of age and older with moderate-to-severe atopic dermatitis as defined by Investigator's Global Assessment (IGA) score ≥ 3 , Eczema Area and Severity Index (EASI) score ≥ 16 , body surface area (BSA) involvement $\geq 10\%$, and Peak Pruritus Numerical Rating Scale (PP-NRS) ≥ 4 at the baseline visit prior to randomization. Patients who had a prior inadequate response or for whom topical treatments were medically inadvisable, or who had received systemic therapies were eligible for inclusion.

All patients who completed the parent studies were eligible to enroll into the long-term extension study EXTEND.

Clinical response

Treatment with CIBINQO 100 mg or 200 mg once daily as monotherapy or in combination with background medicated topical therapy resulted in improvement in objective signs of atopic dermatitis and patient-reported pruritus.

Baseline characteristics

In the placebo-controlled studies (MONO-1, MONO-2, COMPARE) and the open label induction, randomized withdrawal study (REGIMEN) across all treatment groups 41.4% to 51.1% were female, 59.3% to 77.8% were Caucasian, 15.0% to 33.0% were Asian and 4.1% to 8.3% were Black, and the mean age was 32.1 to 37.7 years. In these studies, 32.2% to 40.8% had a baseline IGA of 4 (severe atopic dermatitis), and 41.4% to 59.5% of patients had received prior systemic treatment for atopic dermatitis. The baseline mean EASI score ranged from 28.5 to 30.9, the baseline PP-NRS ranged from 7.0 to 7.3 and the baseline Dermatology Life Quality Index (DLQI) ranged from 14.4 to 16.0.

Monotherapy studies

In both pivotal monotherapy studies (MONO-1, MONO-2), the proportion of patients who achieved IGA and/or EASI-75 response was significantly higher in patients who received CIBINQO 100 mg or 200 mg once daily compared with placebo at Week 12 (see Table 3).

A significantly higher proportion of patients who achieved PP-NRS4 (defined as an improvement of ≥ 4 points in the severity of PP-NRS) with CIBINQO 100 mg or 200 mg once daily compared with placebo was observed as soon as Week 2 and persisting through

Week 12. Higher proportions of patients achieved PP-NRS4 with CIBINQO 100 mg or 200 mg once daily compared with placebo by Day 6 and Day 3 (2 days after the first dose), respectively (see Table 3).

Table 3. Efficacy results of CIBINQO monotherapy at Week 12

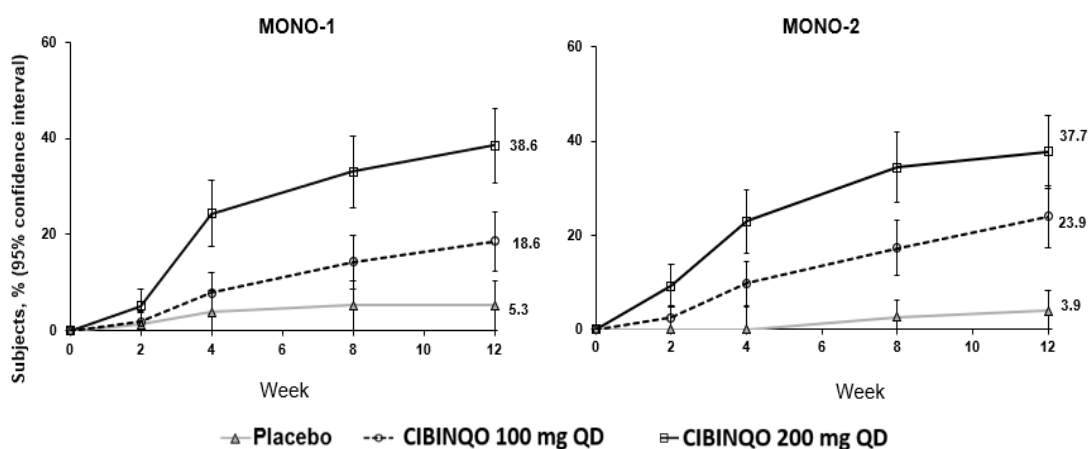
	MONO-1			MONO-2		
	CBQ		Placebo N=77	CBQ		Placebo N=78
	200 mg QD N=154	100 mg QD N=156		200 mg QD N=155	100 mg QD N=158	
	% Responders (95% CI)					
IGA 0 or 1 ^a	43.8 ^g (35.9, 51.7)	23.7 ^e (17.0, 30.4)	7.9 (1.8, 14.0)	38.1 ^g (30.4, 45.7)	28.4 ^f (21.3, 35.5)	9.1 (2.7, 15.5)
EASI-50 ^b	75.8 ^k (69.0, 82.6)	57.7 ^k (49.9, 65.4)	22.4 (13.0, 31.7)	79.9 ^k (73.5, 86.2)	68.4 ^k (61.1, 75.7)	19.5 (10.6, 28.3)
EASI-75 ^b	62.7 ^g (55.1, 70.4)	39.7 ^g (32.1, 47.4)	11.8 (4.6, 19.1)	61.0 ^g (53.3, 68.7)	44.5 ^g (36.7, 52.3)	10.4 (3.6, 17.2)
EASI-90 ^b	38.6 ^k (30.8, 46.3)	18.6 ⁱ (12.5, 24.7)	5.3 (0.2, 10.3)	37.7 ^k (30.0, 45.3)	23.9 ^k (17.2, 30.6)	3.9 (0.0, 8.2)
EASI-100 ^b	13.1 ⁱ (7.7, 18.4)	6.4 ^h (2.6, 10.3)	0 (0.0, 4.7)	7.1 ^h (3.1, 11.2)	5.2 ^h (1.7, 8.6)	0 (0.0, 4.7)
PP-NRS4 ^{c,d}	57.2 ^g (48.8, 65.6)	37.7 ^f (29.2, 46.3)	15.3 (6.6, 24.0)	55.3 ^g (47.2, 63.5)	45.2 ^g (37.1, 53.3)	11.5 (4.1, 19.0)
PP-NRS (0 or 1)	35.4 ^k (27.2, 43.6)	21.1 ⁱ (13.9, 28.4)	3.2 (0.0, 7.5)	32.4 ^k (24.5, 40.2)	21.3 ⁱ (14.5, 28.0)	5.5 (0.3, 10.7)
	% Change from baseline (95% CI)					
LSM EASI	-73.5 ^k (-79.1, -68.0)	-57.5 ^k (-63.1, -51.9)	-28.4 (-36.5, -20.3)	-73.3 ^k (-79.7, -66.9)	-60.0 ^k (-66.5, -53.6)	-28.6 (-38.4, -18.8)
LSM PP-NRS	-56.5 ^k (-63.6, -49.5)	-39.5 ⁱ (-46.7, -32.3)	-19.5 (-30.0, -9.0)	-56.9 ^k (-64.0, -49.8)	-43.5 ⁱ (-50.7, -36.3)	-20.8 (-31.6, -9.9)
LSM SCORAD	-55.1 ^k (-60.1, -50.2)	-41.5 ^k (-46.5, -36.5)	-21.6 (-28.7, -14.5)	-56.2 ^k (-61.2, -51.1)	-45.8 ^k (-50.9, -40.7)	-22.7 (-30.4, -15.1)
	Change from baseline (95% CI)					
LSM PSAAD	-3.2 ^g (-3.6, -2.8)	-2.2 ^e (-2.6, -1.9)	-1.1 (-1.7, -0.6)	-3.0 ^g (-3.3, -2.7)	-2.4 ^g (-2.8, -2.1)	-0.8 (-1.3, -0.3)

Abbreviations: CBQ=CIBINQO; CI=confidence interval; EASI=Eczema Area and Severity Index; LSM=least squares mean; IGA=Investigator Global Assessment; N=number of patients randomized; PP-NRS=Peak Pruritus Numerical Rating Scale; PSAAD=Pruritus and Symptoms Assessment for Atopic Dermatitis; QD=once daily; SCORAD=SCORing Atopic Dermatitis.

- IGA responders were patients with IGA score of clear (0) or almost clear (1) (on a 5-point scale) and a reduction from baseline of ≥ 2 points.
- EASI-50, -75, -90 and -100 responders were patients with $\geq 50\%$, $\geq 75\%$, $\geq 90\%$, and $\geq 100\%$ improvement, respectively in EASI, from baseline.
- The proportion of PP-NRS4 responders was also significantly higher with CIBINQO 200 mg and 100 mg once daily than placebo at Week 2, Week 4, and Week 8 in both MONO-1 and MONO-2.
- PP-NRS4 responders were patients with ≥ 4 -point improvement in PP-NRS from baseline.
- Multiplicity-controlled $p < 0.01$ versus placebo.
- Multiplicity-controlled $p < 0.001$ versus placebo.
- Multiplicity-controlled $p < 0.0001$ versus placebo.
- Nominal $p < 0.05$ versus placebo.
- Nominal $p < 0.01$ versus placebo.
- Nominal $p < 0.001$ versus placebo.
- Nominal $p < 0.0001$ versus placebo.

The proportion of patients who achieved EASI-90 or PP-NRS4 over time in studies MONO-1 and MONO-2 are shown in Figures 1 and 2.

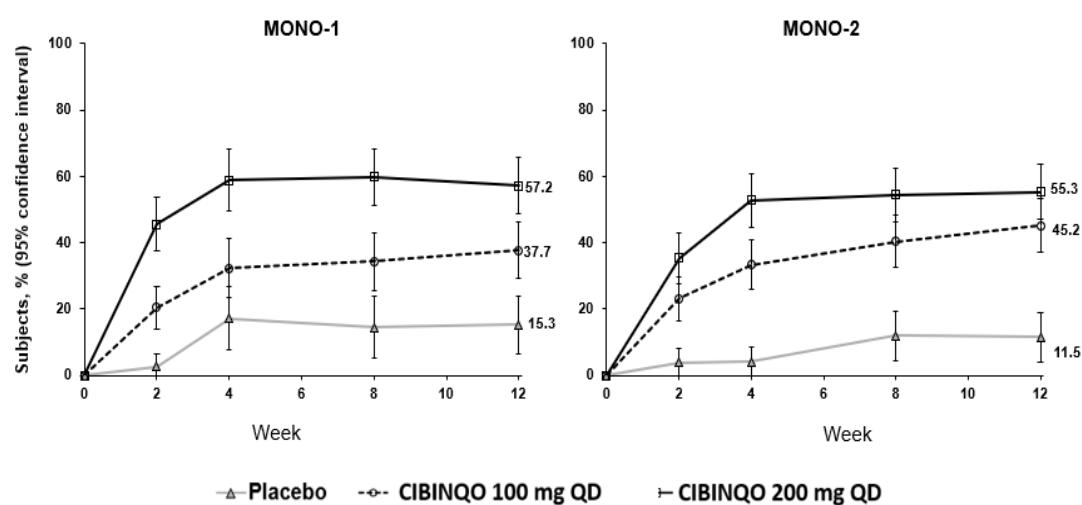
Figure 1. Proportion of patients who achieved EASI-90 over time in MONO-1 and MONO-2



Abbreviations: EASI=Eczema Area and Severity Index; QD=once daily.

PP-NRS4 responders were patients with ≥ 4 -point improvement in Peak Pruritus Numerical Rating Scale (PP-NRS) from baseline.

Figure 2. Proportion of patients who achieved PP-NRS4 over time in MONO-1 and MONO-2



Abbreviations: PP-NRS=Peak Pruritus Numerical Rating Scale; QD=once daily.

PP-NRS4 responders were patients with ≥ 4 -point improvement in Peak Pruritus Numerical Rating Scale (PP NRS) from baseline.

Treatment effects in subgroups (e.g., weight, age, sex, race and prior systemic immunosuppressant treatment) in MONO-1 and MONO-2 were consistent with the results in the overall study population.

Combination therapy study

In the pivotal combination therapy study (COMPARE), the proportion of patients who achieved IGA or EASI-75 response was significantly higher in patients who received CIBINQO 100 mg or 200 mg once daily compared with placebo at Week 12 (see Table 4).

The proportions of patients achieving PP-NRS4 with CIBINQO 100 mg and 200 mg once daily were significantly higher than placebo by Day 9 and Day 4, respectively, and remained significantly higher than placebo with both CIBINQO doses at Week 2 and Week 16.

The proportion of patients achieving PP-NRS4 with CIBINQO 200 mg once daily was significantly higher than dupilumab as early as Day 4 and remained significantly higher than dupilumab at Week 2. The proportion of patients achieving PP-NRS4 was similar between CIBINQO 100 mg once daily and dupilumab at Week 2.

Table 4. Efficacy results of CIBINQO with concomitant topical therapy

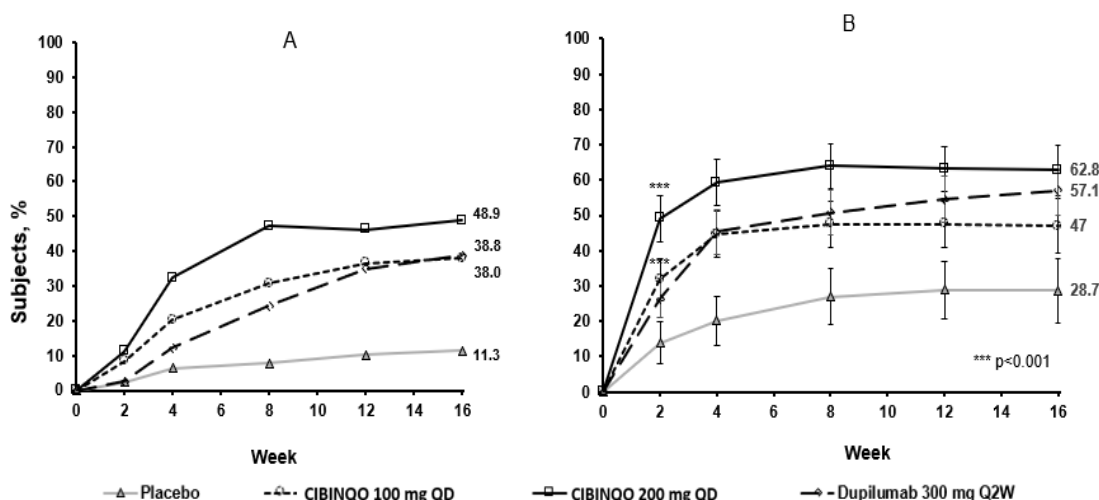
	Week 2				Week 12				Week 16			
	CBQ		PBO N=1 31	DUP N=24 3	CBQ		PBO N=1 31	DUP N=2 43	CBQ		PBO N=1 31	DUP N=2 43
	200 mg N=22 6	100 mg N=23 8			200 mg N=22 6	100 mg N=23 8			200 mg N=22 6	100 mg N=23 8		
	% Responders											
IGA 0 or 1 ^a	18.4 ⁱ	15.2 ^h	6.3	4.7	48.4 ^e	36.6 ^e	14.0	36.5	47.5 ^e	34.8 ^e	12.9	38.8
EASI- 50 ^b	60.5 ^j	53.1 ^j	21.9	35.7	86.3 ^j	75.3 ^j	52.7	80.9	87.3 ^j	81.2 ^j	57.3	84.1
EASI- 75 ^b	30.0 ^j	25.4 ⁱ	10.9	14.0	70.3 ^e	58.7 ^e	27.1	58.1	71.0 ^e	60.3 ^e	30.6	65.5
EASI- 90 ^b	11.2 ^h	8.3 ^g	2.3	2.6	46.1 ^j	36.6 ^j	10.1	34.9	48.9 ^j	38.0 ^j	11.3	38.8
EASI- 100 ^b	4.5 ^g	1.3	0	0.4	12.3 ⁱ	8.1 ^h	1.6	6.6	13.6 ^h	12.7 ^h	4.0	5.2
PP- NRS4 ^c	49.1 ^{e,f}	31.8 ^d	13.8	26.4	63.1 ^j	47.5 ⁱ	28.9	54.5	62.8 ^j	47.0 ^h	28.7	57.1
PP- NRS (0 or 1)	15.0 ^h	8.9	4.6	4.6	36.9 ^j	21.1 ⁱ	7.4	24.9	32.0 ⁱ	24.7 ^g	11.7	24.2
	% Change from baseline											
LSM EASI	-54.6 ^j	-49.3 ^j	-21.2	-38.8	-80.6 ^j	-73.8 ^j	-47.7	-75.4	-83.2 ^j	-75.2 ^j	-53.8	-80.2
LSM PP- NRS	-45.6 ^j	-35.5 ^j	-19.5	-29.3	-63.3 ^j	-48.2 ^j	-30.4	-54.8	-64.1 ^j	-49.1 ^j	-30.3	-58.5
LSM SCOR AD	-41.7 ^j	-34.6 ^j	-18.1	-27.7	-65.2 ^j	-54.2 ^j	-33.5	-58.4	-65.4 ^j	-55.6 ^j	-38.8	-61.9
	Change from baseline											
LSM PSAAD	-2.3 ^j	-1.8 ^j	-0.9	-1.6	-3.6 ^j	-2.7 ^j	-1.6	-3.2	-3.6 ^j	-2.8 ^j	-1.7	-3.4

Abbreviations: CBQ=CIBINQO; DUP=dupilumab; EASI=Eczema Area and Severity Index; LSM=least squares mean; N=number of patients randomized; PBO=placebo; PP-NRS=Peak Pruritus Numerical Rating Scale; PSAAD=Pruritus and Symptoms Assessment for Atopic Dermatitis; SCORAD=SCORing Atopic Dermatitis.

- a. IGA responders were patients with IGA score of clear (0) or almost clear (1) (on a 5-point scale) and a reduction from baseline of ≥ 2 points.
- b. EASI-50, -75, -90 and -100 responders were patients with $\geq 50\%$, $\geq 75\%$, $\geq 90\%$ and $\geq 100\%$ improvement in EASI, respectively, from baseline.
- c. PP-NRS4 responders were patients with ≥ 4 -point improvement in PP-NRS from baseline.
- d. Multiplicity-controlled $p < 0.001$ vs. placebo
- e. Multiplicity-controlled $p < 0.0001$ vs. placebo
- f. Multiplicity-controlled $p < 0.0001$ vs. dupilumab. Statistical comparison between either abrocitinib dose and dupilumab was only performed on the proportion of patients achieving PP-NRS4 at Week 2.
- g. Nominal $p < 0.05$ vs. placebo
- h. Nominal $p < 0.01$ vs. placebo
- i. Nominal $p < 0.001$ vs. placebo
- j. Nominal $p < 0.0001$ vs. placebo

The proportion of patients who achieved EASI-90 or PP-NRS4 over time in COMPARE are shown in Figure 3.

Figure 3. Proportion of patients who achieved A) EASI-90 and B) PP-NRS4 over time in COMPARE



Abbreviations: EASI=Eczema Area and Severity Index; PP-NRS=Peak Pruritus Numerical Rating Scale; QD=once daily; Q2W=every 2 weeks.

EASI-90 was based on EASI $\geq 90\%$ improvement from baseline.

PP-NRS4 response was based on achieving at least 4 points improvement in the severity of Peak Pruritus Numerical Rating Scale (PP-NRS).

Patients who received dupilumab and subsequently enrolled in EXTEND were randomized to either CIBINQO 100 mg or 200 mg once daily upon entering EXTEND. Among responders to dupilumab in COMPARE, the majority maintained response 12 weeks after switching to CIBINQO [77% and 86% for IGA (0 or 1) response, and 90% and 96% for EASI-75 with 100 mg once daily or 200 mg once daily, respectively]. Among non-responders to dupilumab in COMPARE, a substantial proportion of patients achieved response 12 weeks after switching to CIBINQO [34% and 47% for IGA (0 or 1) response, and 68% and 80% for EASI-75 with 100 mg once daily or 200 mg once daily, respectively].

Treatment effects in subgroups (e.g., weight, age, sex, race, and prior systemic immunosuppressant treatment) in COMPARE were consistent with the results in the overall study population.

Late-onset efficacy

Eligible patients who completed the full treatment period of a qualifying parent study (e.g., MONO-1, MONO-2, COMPARE) were considered for enrollment in the long-term extension study EXTEND, which allows patients to extend CIBINQO treatment for at least 92 weeks or until availability of commercial product in their country. In EXTEND, patients received CIBINQO with or without background medicated topical therapy. Patients who were previously randomized to CIBINQO 100 mg or 200 mg once daily in qualifying studies continued the same dose in EXTEND as in the parent study, and the blind was maintained. Patients not previously randomized to CIBINQO in a qualifying parent study were randomized to either CIBINQO 100 mg or 200 mg once daily upon entering EXTEND.

Among patients who did not achieve IGA (0 or 1) response after 12 weeks of CIBINQO treatment and entered EXTEND, 14% and 25% of patients continuing CIBINQO 100 mg once daily in EXTEND achieved IGA (0 or 1) response by Week 16 and Week 24 (with 4 and 12 additional weeks of treatment), respectively, and 19% and 29% of patients continuing CIBINQO 200 mg once daily achieved IGA response by Week 16 and Week 24, respectively (based on observed data). Among patients who did not achieve EASI-75 after 12 weeks of CIBINQO treatment and entered EXTEND, 32% and 50% of patients continuing CIBINQO

100 mg once daily in EXTEND achieved EASI-75 by Week 16 and Week 24 (with 4 and 12 additional weeks of treatment), respectively, and 33% and 57% of patients continuing CIBINQO 200 mg once daily achieved EASI-75 response by Week 16 and Week 24, respectively (based on observed data).

Patients who received dupilumab in the COMPARE study and subsequently entered EXTEND were randomized to either 100 mg or 200 mg of abrocitinib once daily upon entering EXTEND. Among non-responders to dupilumab, a substantial proportion of patients achieved response 12 weeks after switching to abrocitinib [34% and 47% for IGA (0 or 1) response, and 68% and 80% for EASI-75 with 100 mg once daily or 200 mg once daily, respectively].

Long-term efficacy

Among patients who achieved response at Week 12 of a qualifying parent study and entered EXTEND, the majority of patients maintained their response at Week 96 of cumulative CIBINQO treatment for both doses of CIBINQO [64% and 72% for IGA (0 or 1) response, 87% and 90% for EASI-75, and 75% and 80% for PP-NRS4 with 100 mg once daily and 200 mg once daily, respectively (based on observed data)].

Health related outcomes

Treatment with either dose of CIBINQO as monotherapy resulted in significantly improved patient-reported outcomes at 12 weeks compared with placebo (see Table 5). A significantly larger proportion of the CIBINQO groups had clinically meaningful reductions in Dermatology Life Quality Index (DLQI) total scores (defined as a 4-point improvement) from baseline to Week 12 compared with placebo. CIBINQO groups also had a significantly larger proportion of patients who reported “no effect” of their disease on their quality of life (as measured by a DLQI score of 0 or 1).

Both groups significantly improved patient-reported atopic dermatitis symptoms and sleep disruption as measured by the Patient Oriented Eczema Measure (POEM), Night Time Itch Scale (NTIS), and SCORing Atopic Dermatitis (SCORAD) sleep loss subscale. In addition, anxiety and depression symptoms as measured by the Hospital Anxiety and Depression Scale (HADS) total score were significantly reduced in the CIBINQO groups compared with placebo at 12 weeks.

Table 5. Additional endpoint results with CIBINQO monotherapy at Week 12

	MONO-1			MONO-2		
	CBQ		Placebo N=77	CBQ		Placebo N=78
	200 mg QD N=154	100 mg QD N=156		200 mg QD N=155	100 mg QD N=158	
LSM SCORAD (sleep loss subscale)						
Baseline median (SD)	5.9	6.0	6.5	6.2	6.2	5.7
Change from baseline (95% CI)	-3.7 ^d (-4.2, -3.3)	-2.9 ^c (-3.4, -2.5)	-1.6 (-2.2, -1.0)	-3.8 ^d (-4.2, -3.4)	-3.0 ^a (-3.4, -2.6)	-2.1 (-2.7, -1.5)
NTIS >4-point improvement						
% responders	n/a	n/a	n/a	57.0 ^d	42.7 ^d	12.7
DLQI						
0 or 1, % responders	31.9 ^b	20.2	12.1	26.6 ^c	20.3 ^b	5.7
≥4 point improvement, % responders	72.6 ^c	67.2 ^b	43.6	78.1 ^d	73.3 ^d	32.3
LSM DLQI						

Table 5. Additional endpoint results with CIBINQO monotherapy at Week 12

	MONO-1			MONO-2		
	CBQ		Placebo N=77	CBQ		Placebo N=78
	200 mg QD N=154	100 mg QD N=156		200 mg QD N=155	100 mg QD N=158	
Baseline mean (SD)	14.6 (6.8)	14.6 (6.5)	13.9 (7.3)	14.8 (6.0)	15.4 (7.3)	15.0 (7.1)
Change from baseline (95% CI)	-9.1 ^d (-10.3, - 8.0)	-7.0 ^b (-8.1, -5.8)	-4.2 (-5.9, -2.5)	-9.8 ^d (-10.7, -8.8)	-8.3 ^d (-9.3, - 7.3)	-3.9 (-5.3, - 2.4)
CDLQI ≥2.5 point improvement, % responders	83.9 ^a	73.3	53.3	93.3 ^c	56.3 ^a	12.5
LSM-CDLQI Baseline mean (SD)	13.2 (5.5)	11.7 (6.6)	13.6 (7.0)	12.9 (5.7)	13.8 (5.8)	10.1 (3.8)
Change from baseline (95% CI)	-7.5 ^a (-8.9, -6.0)	-6.4 (-7.9, -5.0)	-3.9 (-6.1, -1.7)	-9.7 ^b (-12.1, -7.4)	-4.8 (-7.2, - 2.5)	-2.7 (-6.1, 0.8)
LSM POEM Baseline mean (SD)	19.6 (5.9)	19.5 (6.5)	19.9 (6.1)	19.7 (5.7)	20.9 (5.7)	19.2 (5.5)
Change from baseline (95% CI)	-10.6 ^d (-11.8, - 9.4)	-6.8 ^b (-8.0, -5.6)	-3.7 (-5.5, -1.9)	-11.0 ^d (-12.1, -9.8)	-8.7 ^d (-9.9, - 7.5)	-3.6 (-5.3, - 1.9)
LSM HADS (anxiety) Baseline mean (SD)	5.6 (4.0)	5.9 (4.1)	6.0 (4.0)	5.9 (3.9)	5.5 (4.2)	6.0 (3.7)
Change from baseline (95% CI)	-2.1 ^b (-2.5, -1.6)	-1.6 (-2.0, -1.1)	-1.0 (-1.7, -0.4)	-1.7 ^a (-2.2, -1.2)	-1.6 ^a (-2.1, - 1.1)	-0.6 (-1.3, 0.2)
LSM HADS (depression) Baseline mean (SD)	4.2 (3.7)	4.1 (3.7)	3.9 (3.5)	4.0 (3.7)	4.1 (4.0)	4.4 (3.3)
Change from baseline (95% CI)	-1.8 ^d (-2.2, -1.4)	-1.4 ^b (-1.8, -0.9)	-0.2 (-0.8, 0.4)	-1.4 ^d (-1.8, -1.0)	-1.0 ^c (-1.5, - 0.6)	0.3 (-0.3, 0.9)

Abbreviations: CBQ= CIBINQO; CDLQI=Child Dermatology Life Quality Index; CI=confidence interval; DLQI=Dermatology Life Quality Index; HADS=Hospital Anxiety and Depression Scale; LSM=least squares mean; N=number of patients randomized; n/a= not available; NTIS=Night Time Itch Scale Severity; POEM=Patient Oriented Eczema Measure; QD=once daily; SCORAD=SCORing Atopic Dermatitis.

- a. Nominal p <0.05 versus placebo.
- b. Nominal p <0.01 versus placebo.
- c. Nominal p <0.001 versus placebo.
- d. Nominal p <0.0001 versus placebo.

In COMPARE, a significantly larger proportion of the CIBINQO groups had clinically meaningful reductions in DLQI total scores (defined as a 4-point improvement) from baseline to Week 12 compared with placebo (see Table 6). CIBINQO groups also had a significantly larger proportion of patients who reported “no effect” of their disease on their quality of life (as measured by a DLQI score of 0 or 1).

Both groups significantly improved patient-reported atopic dermatitis symptoms and sleep disruption as measured by the POEM and SCORAD sleep loss subscale, respectively. In addition, anxiety and depression symptoms as measured by the HADS total score were significantly reduced in the CIBINQO groups compared with placebo at 12 weeks.

Table 6. Additional endpoint results with CIBINQO in combination with medicated topical therapies at Week 12

	COMPARE		
	CBQ		Placebo + Topical N=131
	200 mg QD + Topical N=226	100 mg QD + Topical N=238	
LSM SCORAD (sleep loss subscale)			
Baseline mean values	6.4	6.1	6.0
Change from baseline (95% CI)	-4.6 ^d (-4.9, -4.3)	-3.7 ^d (-4.0, -3.4)	-2.4 (-2.8, -2.0)
NTIS >4-point improvement % responders	64.3 ^d	54.0 ^c	34.4
DLQI			
0 or 1, % responders	29.7% ^d	21.9% ^b	8.6%
≥4 point improvement, % responders	86.4% ^d	74.7% ^c	56.5%
LSM DLQI			
Baseline mean (SD)	16.3 (6.6)	15.5 (6.4)	15.2 (6.9)
Change from baseline (95% CI)	-11.0 ^d (-11.7, -10.3)	-8.7 ^d (-9.4, -8.0)	-6.2 (-7.1, -5.3)
LSM POEM			
Baseline mean (SD)	21.5 (5.3)	20.9 (5.5)	20.4 (6.1)
Change from baseline (95% CI)	-12.6 ^d (-13.6, -11.7)	-9.6 ^d (-10.5, -8.6)	-5.1 (-6.3, -3.9)
LSM HADS (anxiety)			
Baseline mean (SD)	5.5 (3.8)	5.3 (3.9)	5.3 (3.9)
Change from baseline (95% CI)	-1.6 ^c (-2.0, -1.2)	-1.2 ^a (-1.5, -0.8)	-0.4 (-0.9, 0.1)
LSM HADS (depression)			
Baseline mean (SD)	3.9 (3.4)	4.0 (3.3)	4.1 (3.7)
Change from baseline (95% CI)	-1.6 ^d (-1.9, -1.2)	-1.3 ^c (-1.6, -0.9)	-0.3 (-0.7, 0.2)

Abbreviations: CBQ= CIBINQO; DLQI=Dermatology Life Quality Index; HADS=Hospital Anxiety and Depression Scale; LSM=least squares mean; NTIS=Night Time Itch Scale Severity; POEM=Patient Oriented Eczema Measure; QD=once daily; SCORAD=SCORing Atopic Dermatitis; SD=standard deviation.

- a. Nominal p <0.05 versus placebo.
- b. Nominal p <0.01 versus placebo.
- c. Nominal p <0.001 versus placebo.
- d. Nominal p <0.0001 versus placebo.

Open label induction, randomized withdrawal study (REGIMEN)

A total of 1,233 patients received open label CIBINQO. Seven-hundred ninety-eight (798) induction responders were randomized to 200 mg or 100 mg of medicinal product or placebo.

Continuous treatment (200 mg continuous) and induction-maintenance treatment (200 mg for 12 weeks followed by 100 mg) prevented flare with 81.1% and 57.4% probability, respectively, versus 19.1% among patients who withdrew treatment (randomized to placebo) after 12 weeks of induction. Three-hundred fifty-one (351) patients including 16.2% of 200 mg, 39.2% of 100 mg and 76.4% of placebo patients received rescue medication of 200 mg CIBINQO in combination with topical therapy.

Table 7. Efficacy results of Cibinqo in REGIMEN

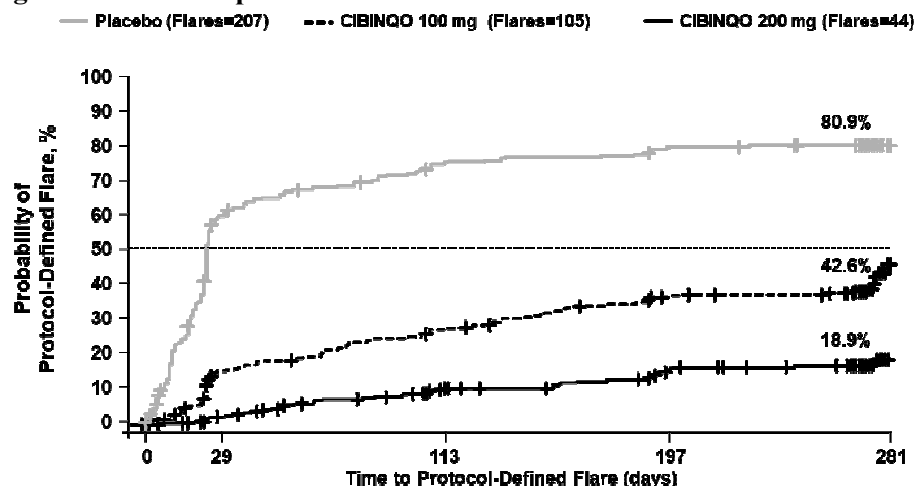
	CBQ monotherapy Open label induction, Week 12 200 mg N=1233
IGA 0 or 1 ^a % responders (95% CI)	65.9 (63.3, 68.6)
EASI-75 ^b % responders (95% CI)	75.6 (73.1, 78.0)

PP-NRS 4-point improvement ^c % responders (95% CI)	68.3 (65.3, 71.3)
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Abbreviations: CBQ=Cibinqo; CI=confidence interval; EASI=Eczema Area and Severity Index; IGA=Investigator Global Assessment; N=number of patients randomized; PP-NRS=Peak Pruritus Numerical Rating Scale.

- IGA responders were patients with IGA score of clear (0) or almost clear (1) (on a 5-point scale) and a reduction from baseline of ≥ 2 points.
- EASI-75 responders were patients with $\geq 75\%$ improvement in EASI from baseline.
- PP-NRS4 responders were patients with ≥ 4 -point improvement in PP-NRS from baseline.

Figure 4. Time to protocol-defined flare



Cibinqo used in monotherapy

Protocol-defined flare=A loss of at least 50% of the EASI response at Week 12 and an IGA score of 2 or higher. Multiplicity-controlled $p < 0.0001$ 200 mg versus placebo; 100 mg versus placebo; 200 mg versus 100 mg.

A multivariate analysis was performed to identify predictors of successfully decreasing the dose from 200 mg to 100 mg and remaining flare-free for at least 12 weeks after the dose decrease. In that analysis, patients who had not received prior systemic agents (OR 1.8, 95% CI: 1.2, 2.6) and patients who had $\leq 50\%$ BSA involvement before starting abrocitinib (OR 1.8, 95% CI: 1.2, 2.6) were almost twice as likely to remain protocol-defined flare-free than those who had received prior systemic agents and who had $> 50\%$ BSA involvement.

Pediatric population

The efficacy and safety of CIBINQO as monotherapy was evaluated in 2 Phase 3 randomized, double-blind, placebo-controlled studies (MONO-1, MONO-2) which included 124 patients who were 12 to less than 18 years of age. The efficacy and safety were also evaluated in open label induction, randomized withdrawal study (REGIMEN) which included 246 patients who were 12 to less than 18 years of age. In these studies, the results in the adolescent subgroup were consistent with the results in the overall study population.

The efficacy and safety of CIBINQO in combination with background medicated topical therapy was evaluated in the Phase 3 randomized, double-blind, placebo-controlled study TEEN. The study included 285 patients who were 12 to less than 18 years of age with moderate-to-severe atopic dermatitis as defined by IGA score ≥ 3 , EASI score ≥ 16 , BSA involvement $\geq 10\%$, and PP NRS ≥ 4 at the baseline visit prior to randomization. Patients who had a prior inadequate response or who had received systemic therapy, were eligible for inclusion.

Baseline characteristics

In TEEN, across all treatment groups 49.1% were female, 56.1% were Caucasian, 33.0% were Asian and 6.0% were Black patients. The median age was 15 years and the proportion of patients with severe atopic dermatitis (IGA of 4) was 38.6%.

Table 8. Adolescent efficacy results of Cibinqo in TEEN

	TEEN ^d		
	CBQ		PBO
	200 mg QD N=94	100 mg QD N=95	
IGA 0 or 1 ^a	46.2 ^e	41.6 ^e	24.5
% responders (95% CI)	(36.1, 56.4)	(31.3, 51.8)	(15.8, 33.2)
EASI-75 ^b	72.0 ^e	68.5 ^e	41.5
% responders (95% CI)	(62.9, 81.2)	(58.9, 78.2)	(31.5, 51.4)
PP-NRS4 ^c	55.4 ^e	52.6	29.8
% responders (95% CI)	(44.1, 66.7)	(41.4, 63.9)	(20.0, 39.5)

Abbreviations: CBQ=Cibinqo; CI=confidence interval; EASI=Eczema Area and Severity Index; IGA=Investigator Global Assessment; N=number of patients treated; PBO=placebo; PP-NRS=Peak Pruritus Numerical Rating Scale; QD=once daily.

- IGA responders were patients with IGA score of clear (0) or almost clear (1) (on a 5-point scale) and a reduction from baseline of ≥ 2 points.
- EASI-75 responders were patients with $\geq 75\%$ improvement in EASI from baseline.
- PP-NRS4 responders were patients with ≥ 4 -point improvement in PP-NRS from baseline.
- Cibinqo used in combination with medicated topical therapy.
- Statistically significant with adjustment for multiplicity versus placebo

Among adolescent patients who achieved response after 12 weeks of treatment and entered long-term extension study EXTEND, the majority of patients maintained their response at Week 96 of cumulative treatment for both doses of abrocitinib [62% and 78%, for IGA (0 or 1) response, 89% and 93% for EASI-75, and 77% and 76% for PP-NRS4] with 100 mg and 200 mg once daily, respectively.

Among adolescent patients who did not achieve response after 12 weeks of treatment and entered EXTEND, a proportion of patients achieved late-onset response by Week 24 (from baseline) of continued treatment with both doses of abrocitinib [34% and 28% for IGA (0 or 1) response, and 41% and 55% for EASI-75] with 100 mg and 200 mg once daily, respectively.

5.2 Pharmacokinetic Properties

Absorption

Effect of Food

Abrocitinib is well-absorbed with over 91% extent of oral absorption and absolute oral bioavailability of approximately 60%. The oral absorption of abrocitinib is rapid and peak plasma concentrations are achieved within 1 hour. Steady-state plasma concentrations of abrocitinib are achieved within 48 hours after once daily administration. Both $C_{\max}$ and AUC of abrocitinib increased dose proportionally up to 400 mg. Coadministration of CIBINQO with a high-fat meal had no clinically relevant effect on abrocitinib exposures (AUC and $C_{\max}$ increased by approximately 26% and 29%, respectively, and $T_{\max}$ was prolonged by 2 hours). In clinical studies, CIBINQO was administered without regard to food (see Section 4.2).

Distribution

After intravenous administration, the volume of distribution of abrocitinib is about 100 L. Approximately 64%, 37% and 29% of circulating abrocitinib and its active metabolites M1 and M2, respectively, are bound to plasma proteins. Abrocitinib and its active metabolites distribute equally between red blood cells and plasma.

Metabolism

The metabolism of abrocitinib is mediated by multiple CYP enzymes, CYP2C19 (~53%), CYP2C9 (~30%), CYP3A4 (~11%) and CYP2B6 (~6%). In a human radiolabeled study, abrocitinib was the most prevalent circulating species, with 3 polar mono-hydroxylated metabolites identified as M1 (3-hydroxypropyl), M2 (2-hydroxypropyl), and M4 (pyrrolidinone pyrimidine). Of the 3 metabolites in circulation, M1 and M2 have similar JAK inhibitory profiles as abrocitinib, while M4 was pharmacologically inactive. The pharmacologic activity of abrocitinib is attributable to the unbound exposures of parent molecule (~60%) as well as M1 (~10%) and M2 (~30%) in systemic circulation. The sum of unbound exposures of abrocitinib, M1 and M2, each expressed in molar units and adjusted for relative potencies, is referred to as the abrocitinib active moiety.

In vitro, abrocitinib or its metabolites were not significant inhibitors or inducers of CYP enzymes (CYP2C8, CYP2C9, and CYP2D6) or of uridine diphosphate glucuronyltransferases (UGTs) (UGT1A1, UGT1A4, UGT1A6, UGT1A9, and UGT2B7). Abrocitinib or its metabolites at clinically meaningful concentrations are not inhibitors of organic anion transporter (OAT)3, organic cation transporter (OCT)1, multidrug and toxin compound extrusion protein (MATE)1/2K and breast cancer resistance protein (BCRP), organic anion transporting polypeptide (OATP) 1B1/1B3, bile salt export pump (BSEP), OAT1 or OCT2

Elimination

The elimination half-life of abrocitinib is about 5 hours. Abrocitinib is eliminated primarily by metabolic clearance mechanisms, with less than 1% of the dose excreted in urine as unchanged drug. The metabolites of abrocitinib, M1, M2 and M4 are excreted predominantly in urine, and are substrates of OAT3 transporter.

Special populations

Body Weight, Gender, Genotype, Race, and Age

Body weight, gender, CYP2C19/2C9 genotype, race, and age did not have a clinically meaningful effect on abrocitinib exposure (see Section 4.2).

Adolescents (12 to less than 18 years of age)

Based on population pharmacokinetic analysis, there was no clinically significant difference in mean abrocitinib steady-state exposures in adolescent patients compared to adults at their typical body weights.

Pediatric (under 12 years of age)

The pharmacokinetics of abrocitinib in pediatric patients under 12 years of age have not yet been established (see Section 4.2).

Renal impairment

In a renal impairment study, patients with severe (eGFR <30 mL/min) and moderate (eGFR 30 to <60 mL/min) renal impairment had approximately 191% and 110% increase in active moiety AUC_{inf}, respectively, compared to patients with normal renal function (eGFR ≥90 mL/min; see Section 4.2). Pharmacokinetics of abrocitinib have not been determined in patients with mild renal impairment, however, based on the results observed in other groups, an increase of up to 70% in active moiety exposure is expected in patients with mild renal impairment (eGFR 60 to <90 mL/min). The increase of up to 70% is not clinically meaningful as the efficacy and safety of abrocitinib in atopic dermatitis patients with mild renal impairment (n=756) was comparable to the overall population in Phase 2 and 3 clinical studies. Based on these results, a clinically significant increase in abrocitinib active moiety is not expected in patients with mild renal impairment (creatinine clearance 60 to <90 mL/min). The eGFR in individual patients was estimated using Modification of Diet in Renal Disease

(MDRD) formula.

CIBINQO has not been studied in patients with ESRD on renal replacement therapy (see Section 4.2). In Phase 3 clinical studies, CIBINQO was not evaluated in patients with atopic dermatitis with baseline creatinine clearance values less than 40 mL/min.

Hepatic impairment

Patients with mild (Child Pugh A) and moderate (Child Pugh B) hepatic impairment had approximately 4% decrease and 15% increase in active moiety AUC_{inf}, respectively, compared to patients with normal hepatic function. These changes are not clinically significant, and no dose adjustment is required in patients with mild or moderate hepatic impairment (see Section 4.2). In clinical studies, CIBINQO was not evaluated in patients with severe (Child Pugh C) hepatic impairment, or in patients screened positive for active hepatitis B or hepatitis C.

5.3 Preclinical Safety Data

Genotoxicity

CIBINQO is not mutagenic in the bacterial mutagenicity assay (Ames assay). Although CIBINQO is aneugenic in the *in vitro* TK6 micronucleus assay, CIBINQO is not aneugenic or clastogenic based on the results of the *in vivo* rat bone marrow micronucleus assay.

Carcinogenicity

No evidence of tumorigenicity was observed in the 6-month Tg.rasH2 mice administered CIBINQO for at oral doses up to 75 mg/kg/day and 60 mg/kg/day in female and male mice, respectively. In the 2-year oral carcinogenicity study, CIBINQO resulted in a statistically higher incidence of benign thymomas in female rats at exposures greater than or equal to 2.8 times the unbound human AUC at the MRHD of 200 mg. No evidence of CIBINQO related tumorigenicity was observed following oral CIBINQO administration in female rats at exposures equal to 0.6 times the unbound human AUC at the MRHD of 200 mg or in male rats at exposures equal to 14 times the unbound human AUC at the MRHD of 200 mg.

Reproductive and developmental toxicity

CIBINQO had no effects on rat male fertility or spermatogenesis at doses up to 70 mg/kg/day at exposures equal to 26 times the unbound human AUC at the MRHD of 200 mg. CIBINQO resulted in effects on rat female fertility (lower fertility index, corpora lutea, and implantation sites) at exposures equal to 29 times the unbound human AUC at the MRHD of 200 mg and higher post implantation loss at exposures greater than or equal to 11 times the unbound human AUC at the MRHD of 200 mg. The effects on female fertility reversed 1 month after cessation of CIBINQO administration. No effects on female fertility were noted at exposures equal to 2 times the unbound human AUC at the MRHD of 200 mg.

No fetal malformations were observed in embryo-fetal development studies in rats or rabbits. In an embryo-fetal development study in pregnant rabbits, oral administration of CIBINQO during gestation Days 7 to 19 had no effects on embryo-fetal survival or fetal morphological development at exposures equal to 4 times the unbound human AUC at the MRHD of 200 mg. CIBINQO resulted in increased incidence of delayed ossification of the forelimb phalanges at exposures equal to 4 times the unbound human AUC at the MRHD of 200 mg.

In an embryo-fetal development study in pregnant rats, oral administration of CIBINQO during gestation Days 6 to 17 resulted in increased embryo-fetal lethality at exposures equal to 17 times the unbound human AUC at the MRHD of 200 mg. No embryo-fetal lethality was observed in pregnant rats orally dosed with CIBINQO during organogenesis at exposures

equal to 11 times the unbound human AUC at the MRHD of 200 mg. CIBINQO resulted in increased incidences of skeletal variations of short 13th ribs at exposures greater than or equal to 11 times the unbound human AUC at the MRHD of 200 mg and reduced ventral processes, thickened ribs, and unossified metatarsals at exposures equal to 17 times the unbound human AUC at the MRHD of 200 mg. No skeletal variations were noted in rats at exposures equal to 2.4 times the unbound human AUC at the MRHD of 200 mg.

In a pre- and postnatal development study in pregnant rats, oral administration of CIBINQO during gestation Day 6 through lactation Day 21 resulted in dystocia with prolonged parturition and lower offspring body weights at exposures greater than or equal to 11 times the unbound human AUC at the MRHD of 200 mg and lower postnatal survival at exposures equal to 17 times the unbound human AUC at the MRHD of 200 mg. No maternal or developmental toxicity was observed in either dams or offspring at exposures equal to 2.4 times the unbound human AUC at the MRHD of 200 mg.

Administration of abrocitinib to juvenile rats beginning on postnatal Day 21 and older (comparable to a 2-year-old human and older) was not associated with microscopic or macroscopic bone findings. Administration of abrocitinib to juvenile rats beginning on postnatal Day 10 (comparable to a 3-month-old human infant) resulted in adverse microscopic and macroscopic bone findings, including malrotated paws, fractures, and/or femoral head abnormalities.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Tablet core

Microcrystalline cellulose
Dibasic calcium phosphate anhydrous
Sodium starch glycolate
Magnesium stearate

Film-coat

Hypromellose (E464)
Titanium dioxide (E171)
Lactose monohydrate
Macrogol
Triacetin (E1518)
Iron red oxide (E172)

6.2 Incompatibilities

Not applicable.

6.3 Shelf-life

Please refer expiry date on the outer carton.

6.4 Special Precautions for Storage

Store below 30°C. Keep in original package.

6.5 Nature and Contents of Container

CIBINQO 50 mg film-coated tablets

High-density polyethylene (HDPE) bottle and polypropylene closure containing 14 or 30 film-coated tablets.

Polyvinylidene chloride (PVDC) blister with aluminium foil lidding film containing 7 film-coated tablets. Each pack contains 14, 28 or 91 film-coated tablets.

CIBINQO 100 mg film-coated tablets.

HDPE bottle and polypropylene closure containing 14 or 30 film-coated tablets.

PVDC blister with aluminium foil lidding film containing 7 film-coated tablets. Each pack contains 14, 28 or 91 film-coated tablets.

CIBINQO 200 mg film-coated tablets.

HDPE bottle and polypropylene closure containing 14 or 30 film-coated tablets.

PVDC blister with aluminium foil lidding film containing 7 film-coated tablets. Each pack contains 14, 28 or 91 film-coated tablets.

Some product strengths or pack sizes may not be available in your country.

6.6 Special Precautions for Disposal and Other Handling

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

Pfizer Manufacturing Deutschland GmbH
Mooswaldallee 1
79108 Freiburg, Im Breisgau
Germany

Date of Revision: 06 JAN 2025

CIBINQO-0125